For instance, $[A]_{4\times1} = \begin{bmatrix} x+1 \\ x^2 \\ 3x \\ 4 \end{bmatrix}$ is a column matrix whose entries are real valued functions of real variable x. More generally, $A = [a_{ij}]_{m\times1} = [a_{i1}]_{m\times1}$ is a column matrix of order $m\times1$.

Definition 7.4

A matrix $A = [a_{ij}]_{m\times n}$ is said to be a **zero matrix** or **null matrix** or **void matrix** denoted by O if $a_{ij} = 0$ for all values of $1 \le i \le m$ and $1 \le j \le n$.

For instance, $[0]$, $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ are zero matrices of order 1×1, 3×3 and 2×4 respectively.

A matrix A is said to be a **non-zero matrix** if at least one of the entries of A is non-zero.

Square, Diagonal, Unit, Triangular matrices

Definition 7.5

A matrix in which number of rows is equal to the number of columns, is called a **square matrix**. That is, a matrix of order $n \times n$ is often referred to as a square matrix of order n.

For instance, $A = \begin{bmatrix} a & b & c \\ d & c & f \\ g & h & l \end{bmatrix}$ is a square matrix of order 3.

Definition 7.6

In a square matrix $A = [a_{ij}]_{n\times n}$ of order n, the elements $a_{11}, a_{22}, a_{33}, ..., a_{nn}$ are called the **principal diagonal** or simply the **diagonal** or **main diagonal** or **leading diagonal elements.**

Definition 7.7

A square matrix $A = [a_{ij}]_{n\times n}$ is called a **diagonal matrix** if $a_{ij} = 0$ whenever $i \neq j$.

Thus, in a diagonal matrix all the entries except the entries along the main diagonal are zero. For instance,

$$A = \begin{bmatrix} 2.5 & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & 0.5 \end{bmatrix},\ B = \begin{bmatrix} r & 0 \\ 0 & s \end{bmatrix},\ C = [6],\ \text{and } D = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & \cdots & a_{nn} \end{bmatrix}$$

are diagonal matrices of order 3, 2, 1, and n respectively.

Is a square zero matrix, a diagonal matrix?

Definition 7.8

A diagonal matrix whose entries along the principal diagonal are equal, is called a **Scalar matrix**.

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